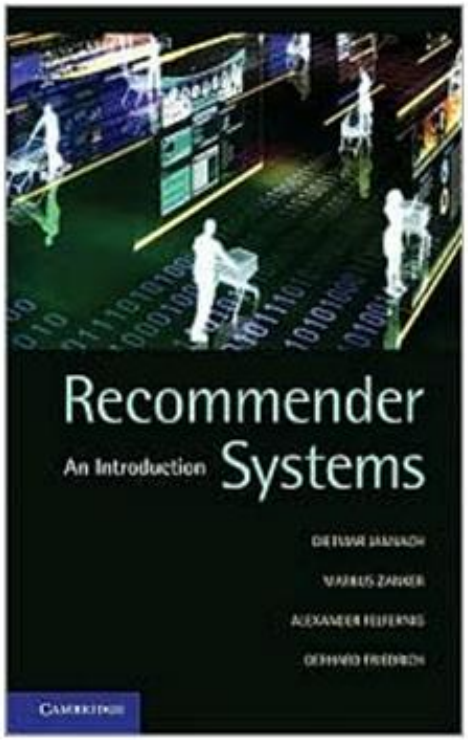
Recommender Systems

Tho Quan
qttho@hcmut.edu.vn



Recommender Systems: An Introduction

by Dietmar Jannach, Markus Zanker, Alexander Felfernig, Gerhard Friedrich

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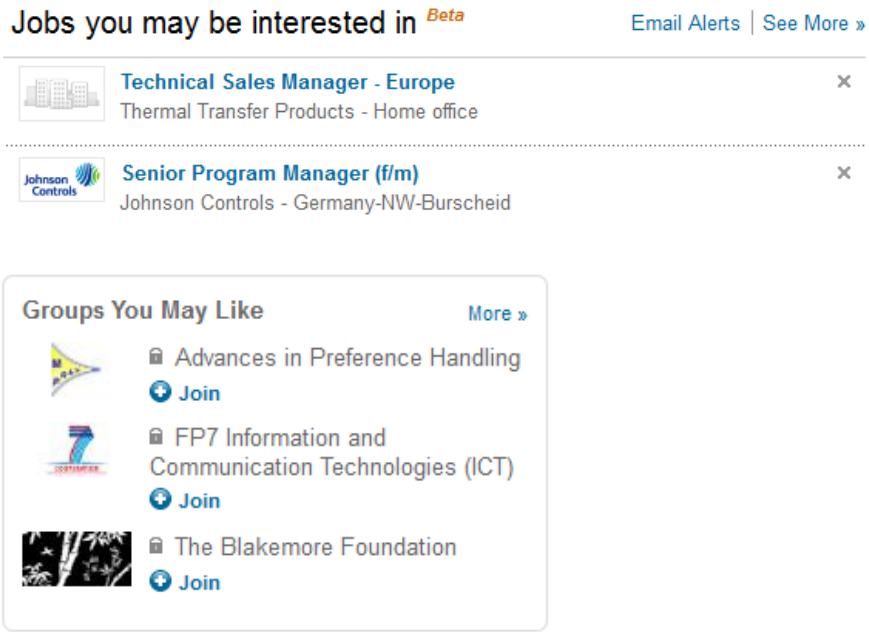
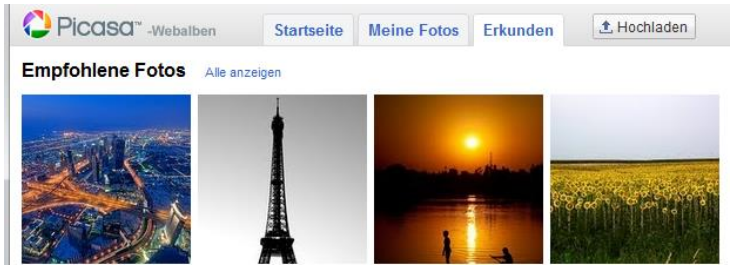
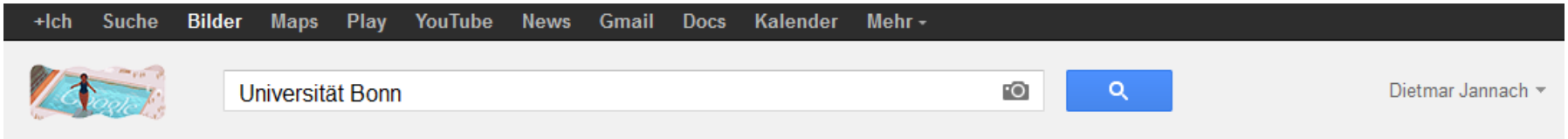




Even more …

- **Personalized search**



- **"Computational advertising"**

Agenda

- **What are recommender systems for?**
  - Introduction
- **How do they work (Part I) ?**
  - Collaborative Filtering
- **How to measure their success?**
  - Evaluation techniques
- **How do they work (Part II) ?**
  - Content-based Filtering
  - Knowledge-Based Recommendations

Introduction

Why using Recommender Systems?

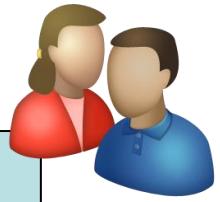
- **Value for the customer**
  - Find things that are interesting
  - Narrow down the set of choices
  - Help me explore the space of options
  - Discover new things
  - Entertainment
  - …
- **Value for the provider**
  - Additional and probably unique personalized service for the customer
  - Increase trust and customer loyalty
  - Increase sales, click trough rates, conversion etc.
  - Opportunities for promotion, persuasion
  - Obtain more knowledge about customers
  - …

Real-world check

- **Myths from industry**
  - Amazon.com generates X percent of their sales through the recommendation lists (30 < X < 70)
  - Netflix (DVD rental and movie streaming) generates X percent of their sales through the recommendation lists (30 < X < 70)
- **There must be some value in it**
  - See recommendation of groups, jobs or people on LinkedIn
  - Friend recommendation and ad personalization on Facebook
  - Song recommendation at last.fm
  - News recommendation at Forbes.com (plus 37% CTR)
- **Academia**
  - A few studies exist that show the effect
    - increased sales, changes in sales behavior

Problem domain

- **Recommendation systems (RS) help to match users with items**
  - Ease information overload
  - Sales assistance (guidance, advisory, persuasion,…)



RS are software agents that elicit the interests and preferences of individual consumers […] and make recommendations accordingly.

They have the potential to support and improve the quality of the decisions consumers make while searching for and selecting products online.

» [Xiao & Benbasat, MISQ, 2007]

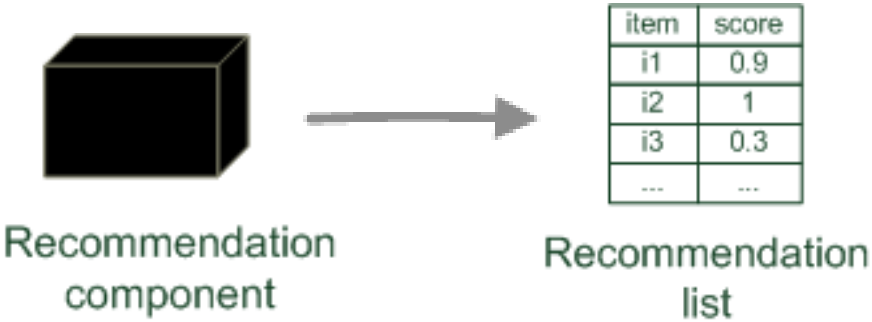
- **Different system designs / paradigms**
  - Based on availability of exploitable data
  - Implicit and explicit user feedback
  - Domain characteristics

Recommender systems

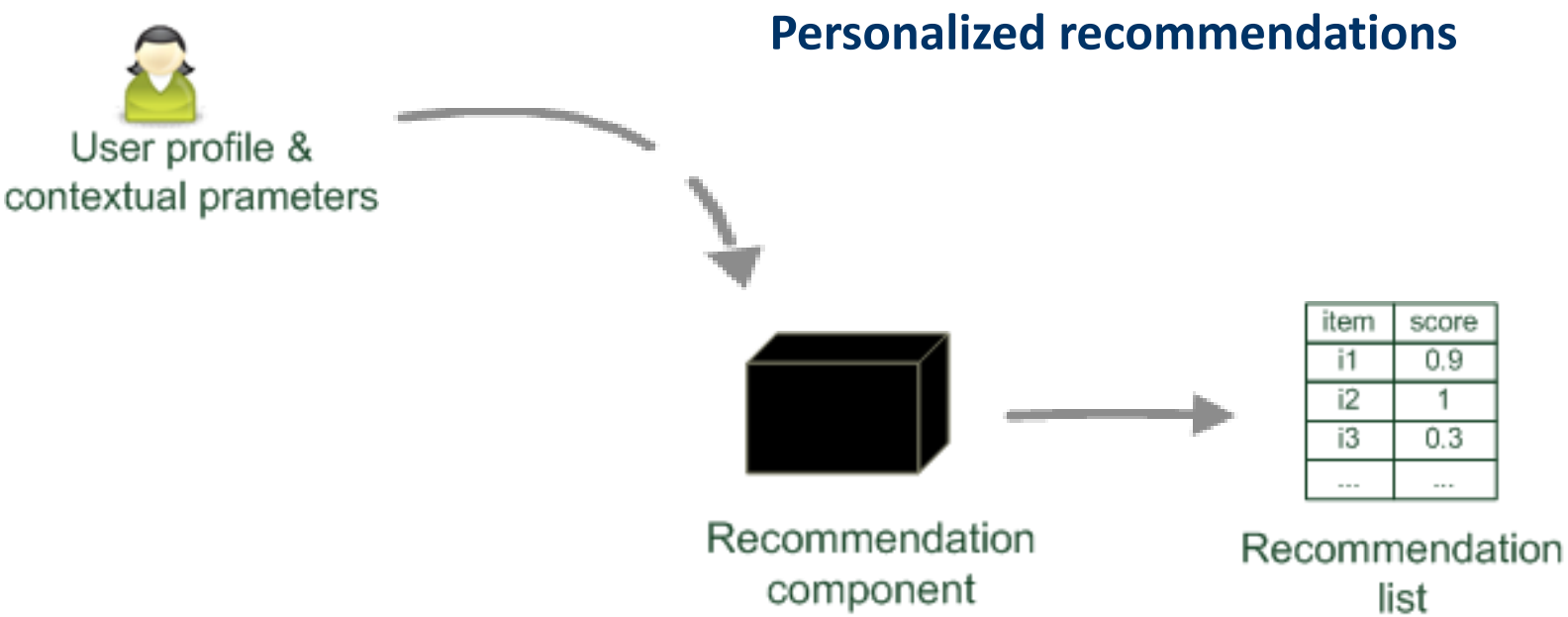
- **RS seen as a function** [AT05]
- **Given:**
  - User model (e.g. ratings, preferences, demographics, situational context)
  - Items (with or without description of item characteristics)
- **Find:**
  - Relevance score. Used for ranking.
- **Finally:**
  - Recommend items that are assumed to be relevant
- **But:**
  - Remember that relevance might be context-dependent
  - Characteristics of the list itself might be important (diversity)

Paradigms of recommender systems

Recommender systems reduce information overload by estimating relevance

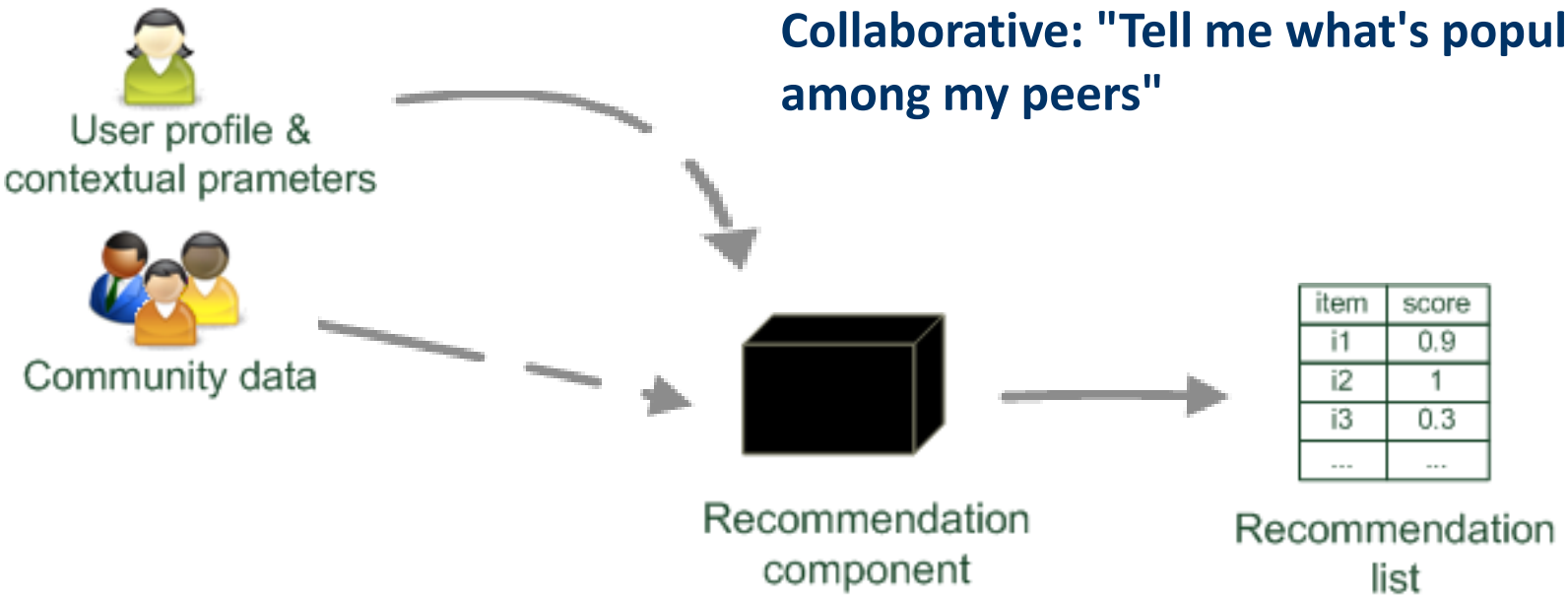


Paradigms of recommender systems



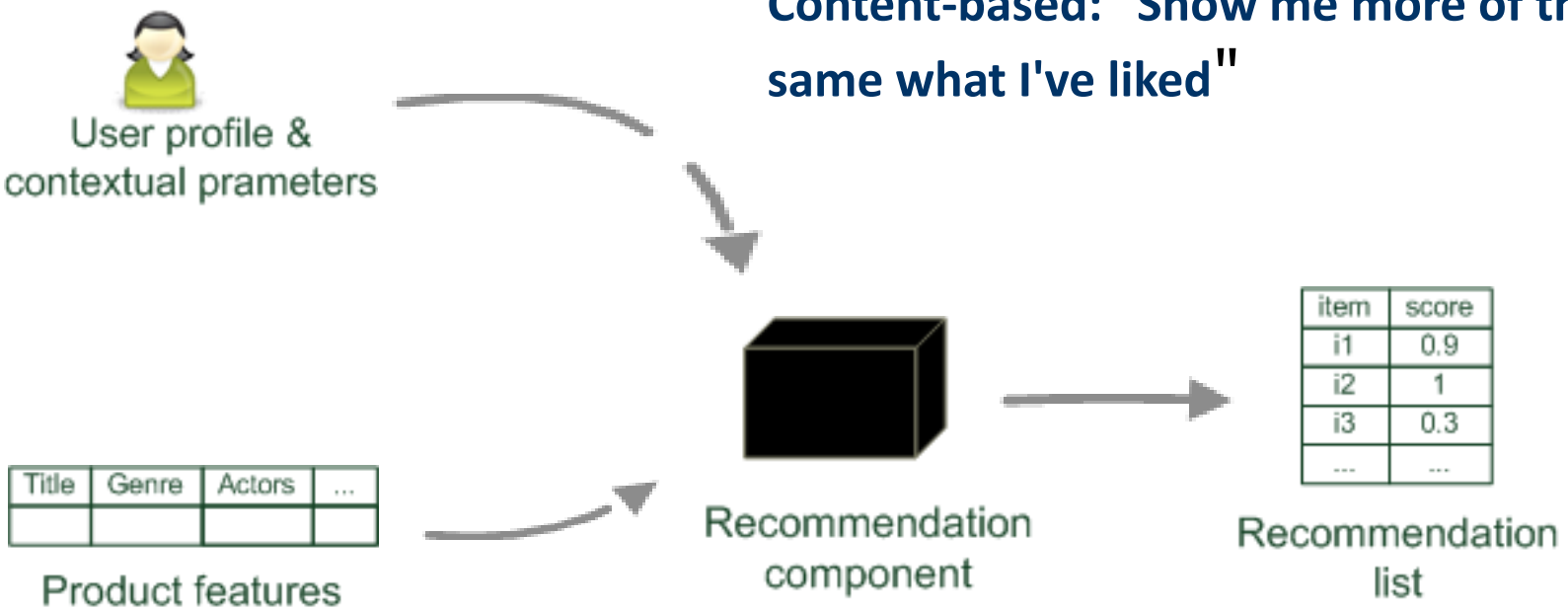
Paradigms of recommender systems

Collaborative: "Tell me what's popular among my peers"



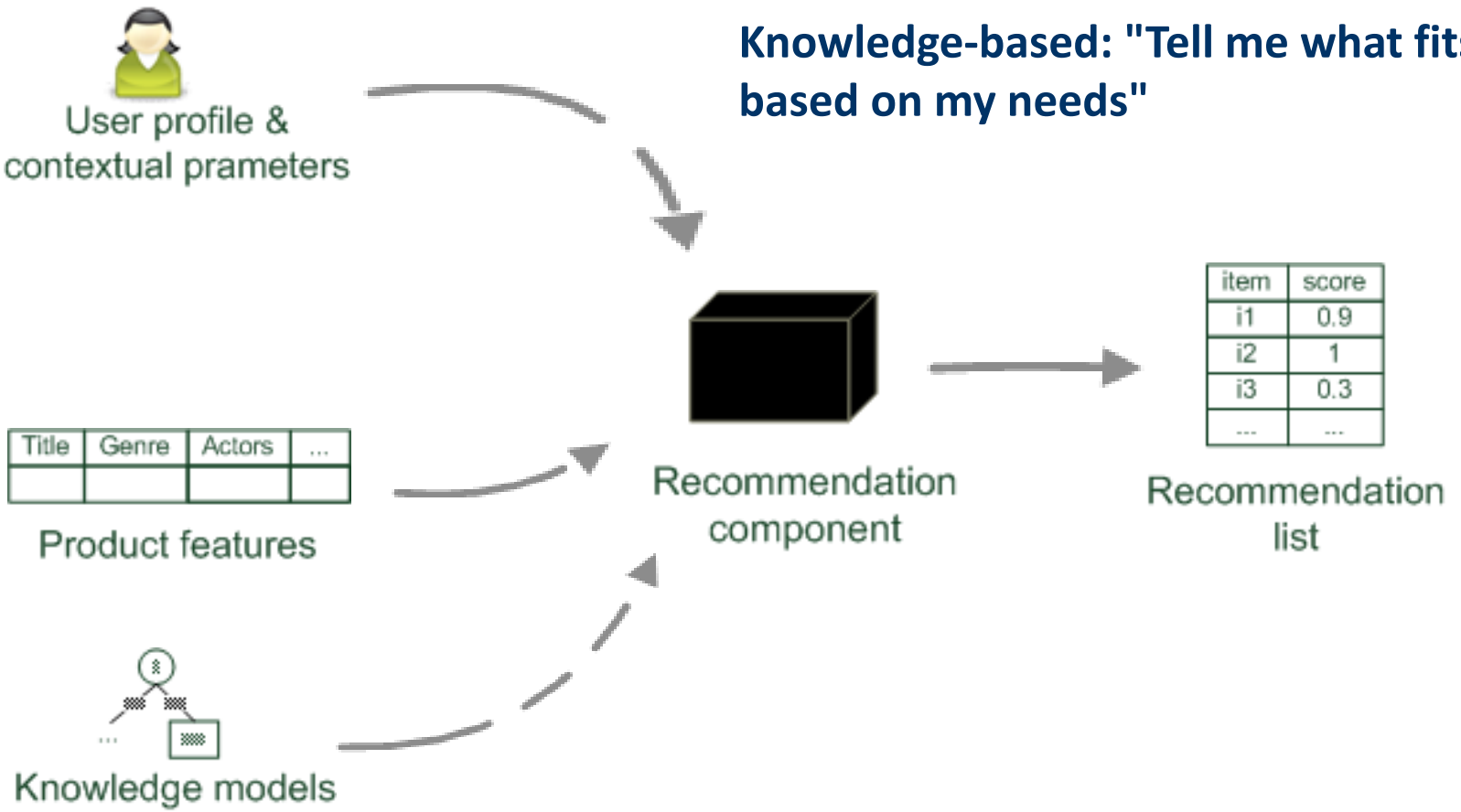
Paradigms of recommender systems

Content-based: "Show me more of the same what I've liked"



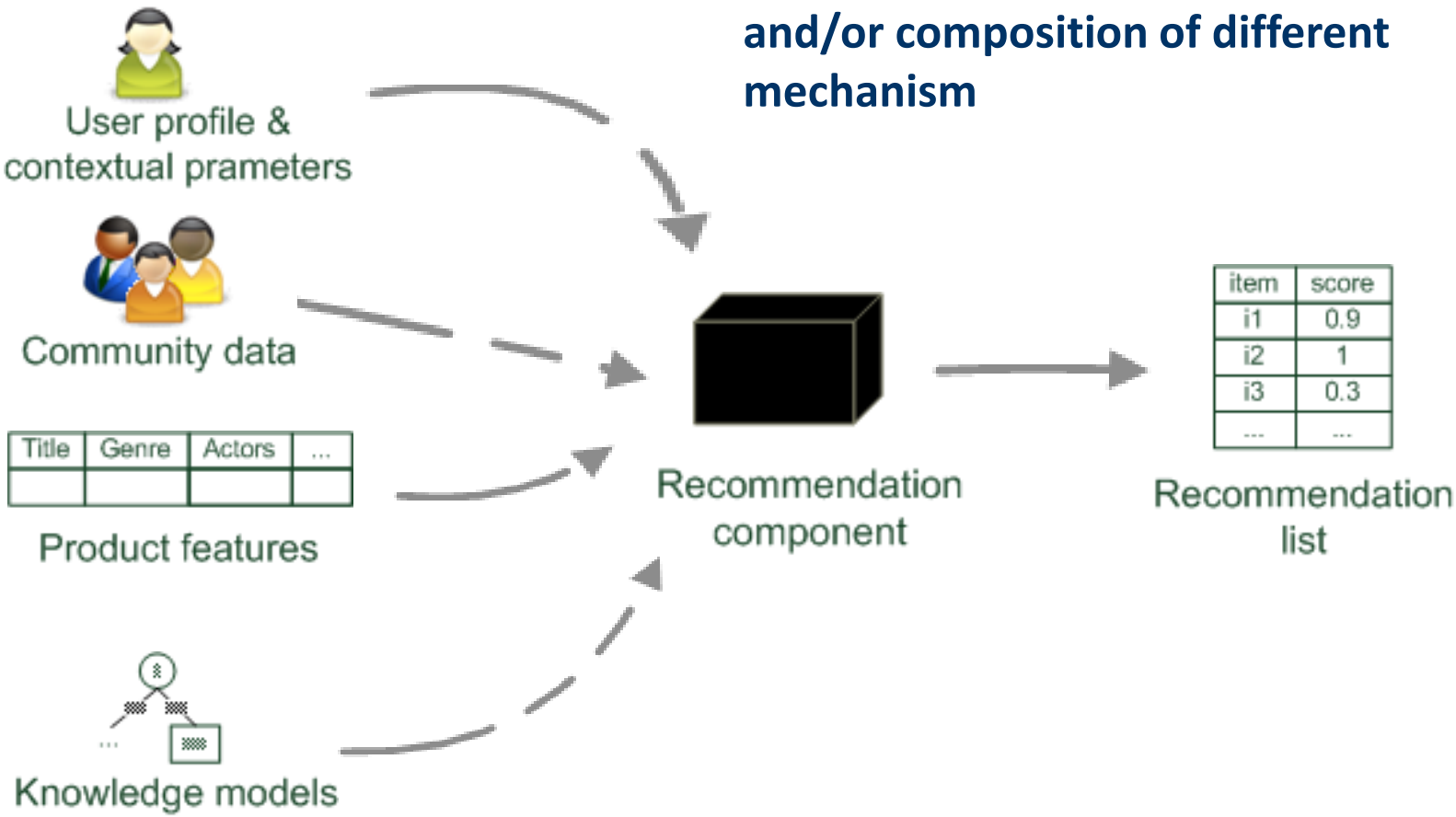
Paradigms of recommender systems

Knowledge-based: "Tell me what fits based on my needs"



Paradigms of recommender systems

Hybrid: combinations of various inputs and/or composition of different mechanism



Recommender systems: basic techniques

	Pros	Cons
Collaborative	No knowledge-engineering effort, serendipity of results, learns market segments	Requires some form of rating feedback, cold start for new users and new items
Content-based	No community required, comparison between items possible	Content descriptions necessary, cold start for new users, no surprises
Knowledge-based	Deterministic recommendations, assured quality, no cold-start, can resemble sales dialogue	Knowledge engineering effort to bootstrap, basically static, does not react to short-term trends

Collaborative Filtering

Collaborative Filtering (CF)

- **The most prominent approach to generate recommendations**
  - used by large, commercial e-commerce sites
  - well-understood, various algorithms and variations exist
  - applicable in many domains (book, movies, DVDs, ..)

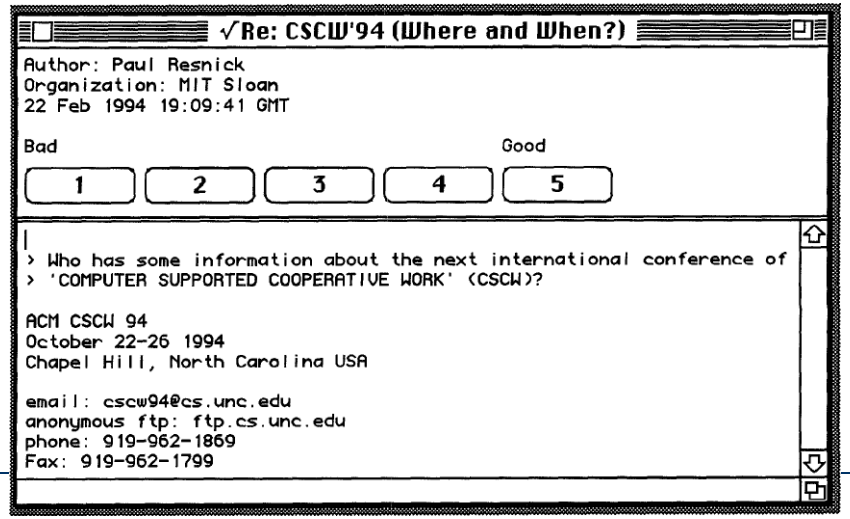
- **Approach**
  - use the "wisdom of the crowd" to recommend items
- **Basic assumption and idea**
  - Users give ratings to catalog items (implicitly or explicitly)
  - Customers who had similar tastes in the past, will have similar tastes in the future

1992: *Using collaborative filtering to weave an information tapestry*, D. Goldberg et al., Communications of the ACM

- **Basic idea: "Eager readers read all docs immediately, casual readers wait for the eager readers to annotate"**
- **Experimental mail system at Xerox Parc that records reactions of users when reading a mail**
- **Users are provided with personalized mailing list filters instead of being forced to subscribe**
  - Content-based filters (topics, from/to/subject…)
  - Collaborative filters
- **E.g. Mails to [all] which were replied by [John Doe] and which received positive ratings from [X] and [Y].**

1994: *GroupLens: an open architecture for collaborative filtering of netnews*, P. Resnick et al., ACM CSCW

- Tapestry system does not aggregate ratings and requires knowing each other
- Basic idea: "People who agreed in their subjective evaluations in the past are likely to agree again in the future"
- Builds on newsgroup browsers with rating functionality



User-based nearest-neighbor collaborative filtering (1)

- **The basic technique:**
  - Given an "active user" (Alice) and an item I not yet seen by Alice
  - The *goal is to estimate Alice's rating for this item*, e.g., by
    - find a set of users (peers) who liked the same items as Alice in the past **and** who have rated item I
    - use, e.g. the average of their ratings to predict, if Alice will like item I
    - do this for all items Alice has not seen and recommend the best-rated

	Item1	Item2	Item3	Item4	Item5
Alice	5	3	4	4	?
User1	3	1	2	3	3
User2	4	3	4	3	5
User3	3	3	1	5	4
User4	1	5	5	2	1

User-based nearest-neighbor collaborative filtering (2)

- **Some first questions**
  - How do we measure similarity?
  - How many neighbors should we consider?
  - How do we generate a prediction from the neighbors' ratings?

	Item1	Item2	Item3	Item4	Item5
Alice	5	3	4	4	?
User1	3	1	2	3	3
User2	4	3	4	3	5
User3	3	3	1	5	4
User4	1	5	5	2	1

Measuring user similarity

- **A popular similarity measure in user-based CF: Pearson correlation**

$$sim(a,b) = \frac{\sum_{p \in P}(r_{a,p} - \bar{r}_a)(r_{b,p} - \bar{r}_b)}{\sqrt{\sum_{p \in P}(r_{a,p} - \bar{r}_a)^2}\ \sqrt{\sum_{p \in P}(r_{b,p} - \bar{r}_b)^2}}$$

a, b : users

$r_{a,p}$: rating of user a for item p

P : set of items, rated both by a and b

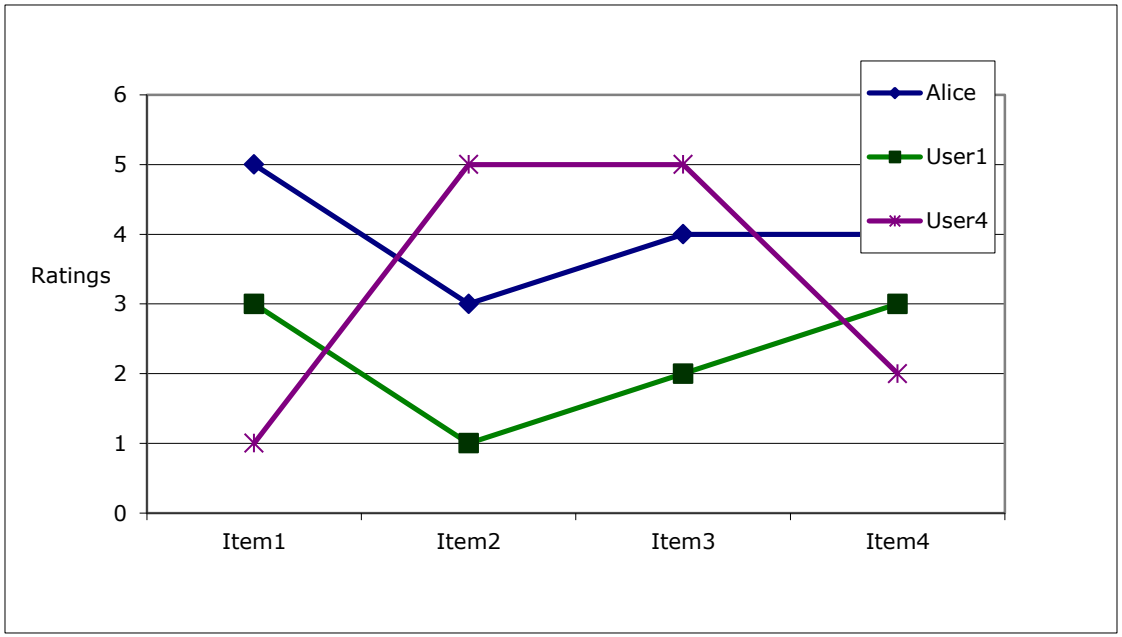
Possible similarity values between -1 and 1; $\overline{r_a}, \overline{r_b}$ = user's average ratings

	Item1	Item2	Item3	Item4	Item5
Alice	5	3	4	4	?
User1	3	1	2	3	3
User2	4	3	4	3	5
User3	3	3	1	5	4
User4	1	5	5	2	1

sim = 0,85
sim = 0,70
sim = -0,79

Pearson correlation

- **Takes differences in rating behavior into account**



- **Works well in usual domains, compared with alternative measures**
  - such as cosine similarity

Making predictions

- A common prediction function:

$$pred(a,p) = \overline{r_a} + \frac{\sum_{b \in N} sim(a,b) * (r_{b,p} - \overline{r_b})}{\sum_{b \in N} sim(a,b)}$$

- Calculate, whether the neighbors' ratings for the unseen item ***i*** are higher or lower than their average
- Combine the rating differences – use the similarity as a weight
- Add/subtract the neighbors' bias from the active user's average and use this as a prediction

Making recommendations

- **Making predictions is typically not the ultimate goal**
- **Usual approach (in academia)**
  - Rank items based on their predicted ratings
- **However**
  - This might lead to the inclusion of (only) niche items
  - **In practice also:** Take item popularity into account
- **Approaches**
  - "Learning to rank"
    - Optimize according to a given rank evaluation metric (see later)

Improving the metrics / prediction function

- **Not all neighbor ratings might be equally "valuable"**
  - Agreement on commonly liked items is not so informative as agreement on controversial items
  - **Possible solution**: Give more weight to items that have a higher variance
- **Value of number of co-rated items**
  - Use "significance weighting", by e.g., linearly reducing the weight when the number of co-rated items is low
- **Case amplification**
  - Intuition: Give more weight to "very similar" neighbors, i.e., where the similarity value is close to 1.
- **Neighborhood selection**
  - Use similarity threshold or fixed number of neighbors

Memory-based and model-based approaches

- **User-based CF is said to be "memory-based"**
  - the rating matrix is directly used to find neighbors / make predictions
  - does not scale for most real-world scenarios
  - large e-commerce sites have tens of millions of customers and millions of items
- **Model-based approaches**
  - based on an offline pre-processing or "model-learning" phase
  - at run-time, only the learned model is used to make predictions
  - models are updated / re-trained periodically
  - large variety of techniques used
  - model-building and updating can be computationally expensive

2001: *Item-based collaborative filtering recommendation algorithms*, B. Sarwar et al., WWW 2001

- **Scalability issues arise with U2U if many more users than items (m >> n , m = |users|, n = |items|)**
  - e.g. Amazon.com
  - Space complexity $O(m^2)$ when pre-computed
  - Time complexity for computing Pearson $O(m^2n)$

- **High sparsity leads to few common ratings between two users**

- **Basic idea: "Item-based CF exploits relationships between items first, instead of relationships between users"**

Item-based collaborative filtering

- **Basic idea:**
  - Use the similarity between items (and not users) to make predictions
- **Example:**
  - Look for items that are similar to Item5
  - Take Alice's ratings for these items to predict the rating for Item5

	Item1	Item2	Item3	Item4	Item5
Alice	5	3	4	4	?
User1	3	1	2	3	3
User2	4	3	4	3	5
User3	3	3	1	5	4
User4	1	5	5	2	1

The cosine similarity measure

- **Produces better results in item-to-item filtering**
  - for some datasets, no consistent picture in literature
- **Ratings are seen as vector in n-dimensional space**
- **Similarity is calculated based on the angle between the vectors**

$$sim(\vec{a}, \vec{b}) = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}| * |\vec{b}|}$$

- **Adjusted cosine similarity**
  - take average user ratings into account, transform the original ratings

$$sim(a, b) = \frac{\sum_{u \in U} (r_{u,a} - \overline{r_u})(r_{u,b} - \overline{r_u})}{\sqrt{\sum_{u \in U} (r_{u,a} - \overline{r_u})^2} \sqrt{\sum_{u \in U} (r_{u,b} - \overline{r_u})^2}}$$

Pre-processing for item-based filtering

- **Item-based filtering does not solve the scalability problem itself**
- **Pre-processing approach by Amazon.com (in 2003)**
  - Calculate all pair-wise item similarities in advance
  - The neighborhood to be used at run-time is typically rather small, because only items are taken into account which the user has rated
  - Item similarities are supposed to be more stable than user similarities
- **Memory requirements**
  - Up to N^2 pair-wise similarities to be memorized (N = number of items) in theory
  - In practice, this is significantly lower (items with no co-ratings)
  - Further reductions possible
    - Minimum threshold for co-ratings (items, which are rated at least by *n* users)
    - Limit the size of the neighborhood (might affect recommendation accuracy)

More on ratings

- **Pure CF-based systems only rely on the rating matrix**
- **Explicit ratings**
  - Most commonly used (1 to 5, 1 to 7 Likert response scales)
  - Research topics
    - "Optimal" granularity of scale; indication that 10-point scale is better accepted in movie domain
    - Multidimensional ratings (multiple ratings per movie)
  - Challenge
    - Users not always willing to rate many items; sparse rating matrices
    - How to stimulate users to rate more items?
- **Implicit ratings**
  - clicks, page views, time spent on some page, demo downloads …
  - Can be used in addition to explicit ones; question of correctness of interpretation

Data sparsity problems

- **Cold start problem**
  - How to recommend new items? What to recommend to new users?
- **Straightforward approaches**
  - Ask/force users to rate a set of items
  - Use another method (e.g., content-based, demographic or simply non-personalized) in the initial phase
- **Alternatives**
  - Use better algorithms (beyond nearest-neighbor approaches)
  - Example:
    - In nearest-neighbor approaches, the set of sufficiently similar neighbors might be to small to make good predictions
    - Assume "transitivity" of neighborhoods

Example algorithms for sparse datasets

- **Recursive CF**
  - Assume there is a very close neighbor ***n*** of u who however has not rated the target item ***i*** yet.
  - Idea:
    - Apply CF-method recursively and predict a rating for item ***i*** for the neighbor
    - Use this predicted rating instead of the rating of a more distant direct neighbor

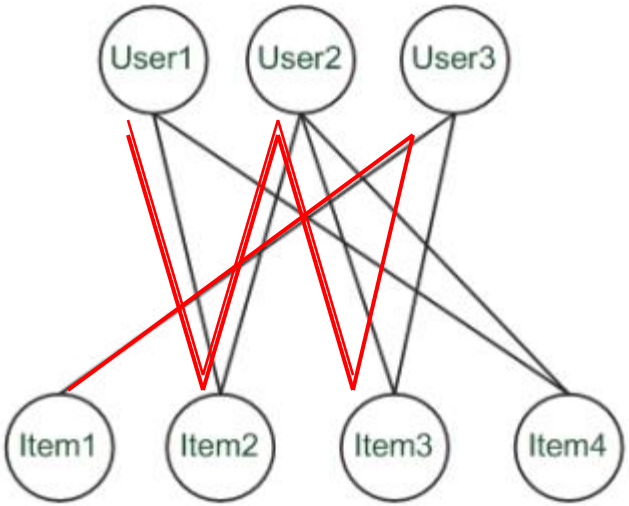
	Item1	Item2	Item3	Item4	Item5
Alice	5	3	4	4	?
User1	3	1	2	3	?
User2	4	3	4	3	5
User3	3	3	1	5	4
User4	1	5	5	2	1

sim = 0,85

Predict rating for User1

Graph-based methods

- **"Spreading activation" (sketch)**
  - Idea: Use paths of lengths > 3 to recommend items
  - Length 3: Recommend Item3 to User1
  - Length 5: Item1 also recommendable



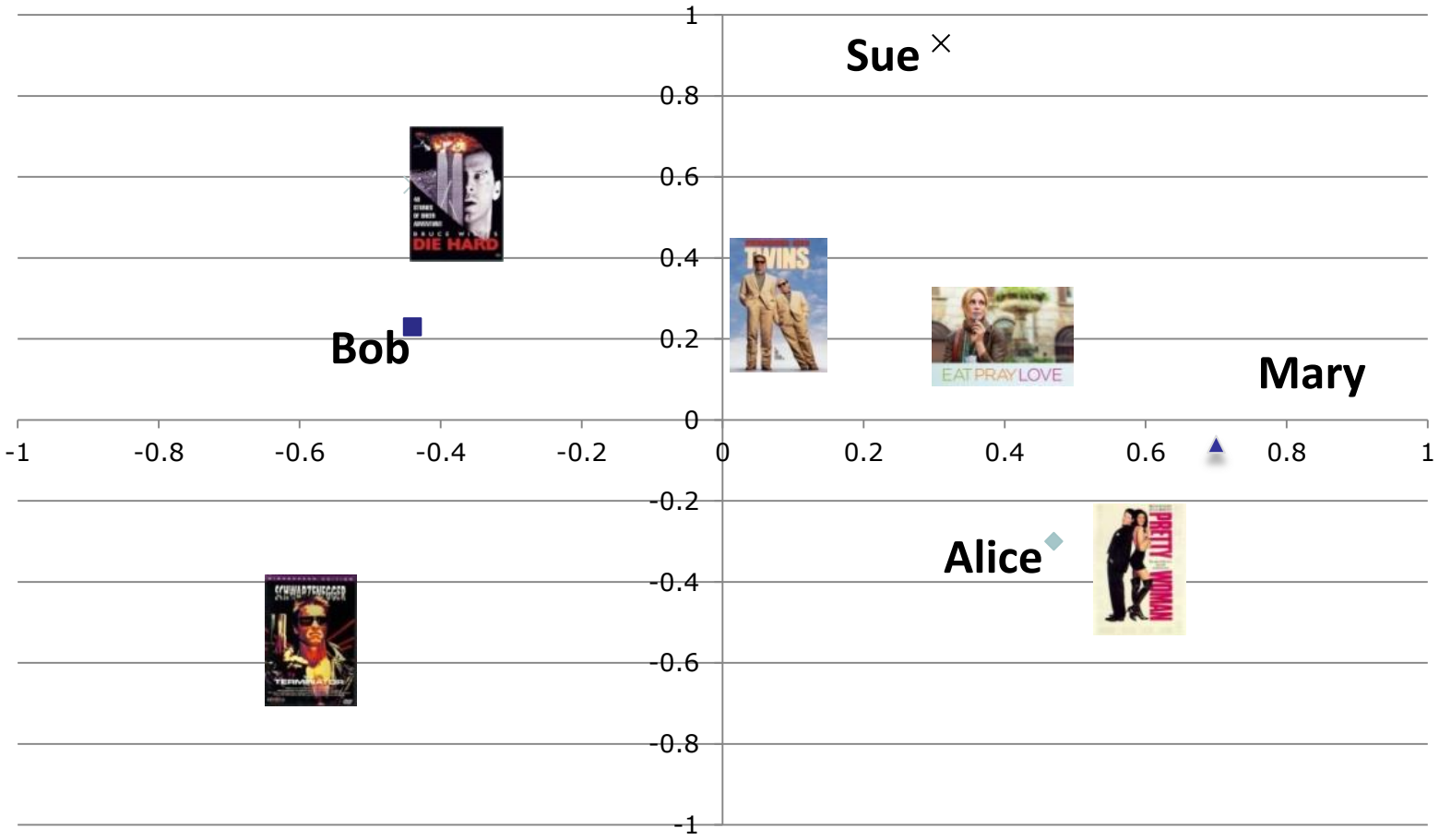
More model-based approaches

- **Plethora of different techniques proposed in the last years, e.g.,**
  - Matrix factorization techniques, statistics
    - singular value decomposition, principal component analysis
  - Association rule mining
    - compare: shopping basket analysis
  - Probabilistic models
    - clustering models, Bayesian networks, probabilistic Latent Semantic Analysis
  - Various other machine learning approaches
- **Costs of pre-processing**
  - Usually not discussed
  - Incremental updates possible?

2000: *Application of Dimensionality Reduction in Recommender System*, B. Sarwar et al., WebKDD Workshop

- **Basic idea: Trade more complex offline model building for faster online prediction generation**
- **Singular Value Decomposition for dimensionality reduction of rating matrices**
  - Captures important factors/aspects and their weights in the data
  - factors can be genre, actors but also non-understandable ones
  - Assumption that k dimensions capture the signals and filter out noise (K = 20 to 100)
- **Constant time to make recommendations**
- **Approach also popular in IR (Latent Semantic Indexing), data compression, …**

A picture says …



Matrix factorization

- **SVD:** $$M_k = U_k \times \Sigma_k \times V_k^T$$

U_k	Dim1	Dim2
Alice	0.47	-0.30
Bob	-0.44	0.23
Mary	0.70	-0.06
Sue	0.31	0.93

V_k^T					
Dim1	-0.44	-0.57	0.06	0.38	0.57
Dim2	0.58	-0.66	0.26	0.18	-0.36

Σ_k	Dim1	Dim2
Dim1	5.63	0
Dim2	0	3.23

- **Prediction:** $$\hat{r}_{ui} = \bar{r}_u + U_k(Alice) \times \Sigma_k \times V_k^T(EPL)$$

= 3 + 0.84 = 3.84

Association rule mining

- **Commonly used for shopping behavior analysis**
  - aims at detection of rules such as

    "If a customer purchases baby-food then he also buys diapers in 70% of the cases"

- **Association rule mining algorithms**
  - can detect rules of the form X => Y (e.g., baby-food => diapers) from a set of sales transactions D = $\{t_1, t_2, \ldots t_n\}$
  - measure of quality: support, confidence

Probabilistic methods

- **Basic idea (simplistic version for illustration):**
  - given the user/item rating matrix
  - determine the probability that user Alice will like an item i
  - base the recommendation on such these probabilities
- **Calculation of rating probabilities based on Bayes Theorem**
  - How probable is rating value "1" for Item5 given Alice's previous ratings?
  - Corresponds to conditional probability P(Item5=1 | X), where
    - X = Alice's previous ratings = (Item1 =1, Item2=3, Item3= …)
  - Can be estimated based on Bayes' Theorem
- **Usually more sophisticated methods used**
  - Clustering
  - pLSA …

2008: *Factorization meets the neighborhood: a multifaceted collaborative filtering model*, Y. Koren, ACM SIGKDD

- **Stimulated by work on Netflix competition**
  - **Prize of $1,000,000 for accuracy improvement of 10% RMSE compared to own Cinematch system**
  - **Very large dataset (~100M ratings, ~480K users , ~18K movies)**
  - **Last ratings/user withheld (set K)**
- **Root mean squared error metric optimized to 0.8567**

$$RMSE = \sqrt{\frac{\sum_{(u,i)\in K} (\hat{r}_{ui} - r_{ui})^2}{|K|}}$$

2008: *Factorization meets the neighborhood: a multifaceted collaborative filtering model*, Y. Koren, ACM SIGKDD

- **Merges neighborhood models with latent factor models**
- **Latent factor models**
  - **good to capture weak signals in the overall data**
- **Neighborhood models**
  - **good at detecting strong relationships between close items**
- **Combination in one prediction single function**
  - Local search method such as stochastic gradient descent to determine parameters
  - Add penalty for high values to avoid over-fitting

$$\hat{r}_{ui} = \mu + b_u + b_i + p_u^T q_i$$

$$\min_{p_*, q_*, b_*} \sum_{(u,i)\in K} (r_{ui} - \mu - b_u - b_i - p_u^T q_i)^2 + \lambda(\|p_u\|^2 + \|q_i\|^2 + b_u^2 + b_i^2)$$

Summarizing recent methods

- **Recommendation is concerned with learning from noisy observations (x, y), where** $f(x) = \hat{y}$

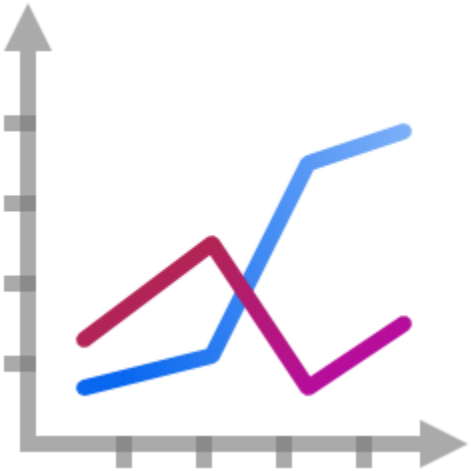
  has to be determined such that $\sum_{\hat{y}} (\hat{y} - y)^2$ **is minimal.**

- **A variety of different learning strategies have been applied trying to estimate f(x)**
  - Non parametric neighborhood models
  - MF models, SVMs, Neural Networks, Bayesian Networks,…

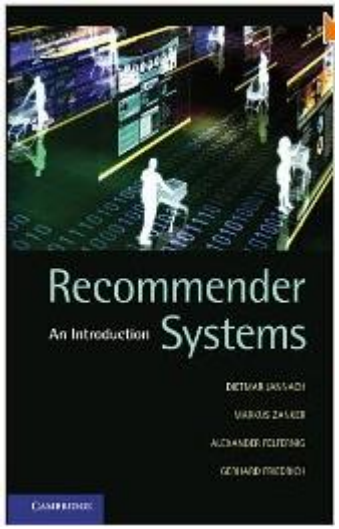
Collaborative Filtering Issues

- **Pros:** 👍
  - well-understood, works well in some domains, no knowledge engineering required
- **Cons:** 👎
  - requires user community, sparsity problems, no integration of other knowledge sources, no explanation of results
- **What is the best CF method?**
  - In which situation and which domain? Inconsistent findings; always the same domains and data sets; differences between methods are often very small (1/100)
- **How to evaluate the prediction quality?**
  - MAE / RMSE: What does an MAE of 0.7 actually mean?
  - Serendipity: Not yet fully understood
- **What about multi-dimensional ratings?**

Evaluation of Recommender Systems



Recommender Systems in e-Commerce



- **One Recommender Systems research question**
  - What should be in that list?

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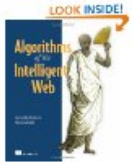
Francesco Ricci
Hardcover
$167.73



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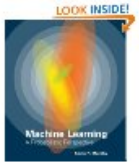

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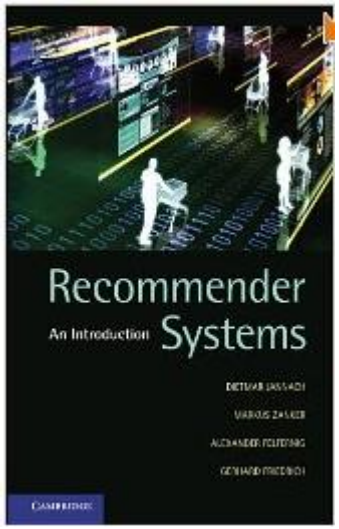

- **Another question both in research and practice**
  - How do we know that these are good recommendations?

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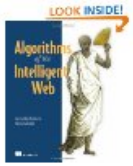

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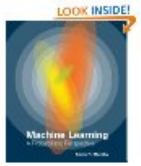

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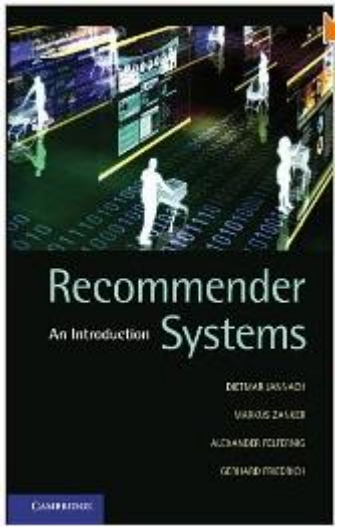

- **This might lead to …**
  - What is a good recommendation?
  - What is a good recommendation strategy?
  - What is a good recommendation strategy for my business?

These have been in stock for quite a while now …

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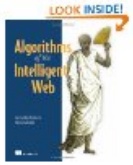

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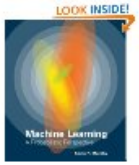

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What is a good recommendation?

What are the measures in practice?

- Total sales numbers
- Promotion of certain items
- …
- Click-through-rates
- Interactivity on platform
- …
- Customer return rates
- Customer satisfaction and loyalty

Purpose and success criteria (1)

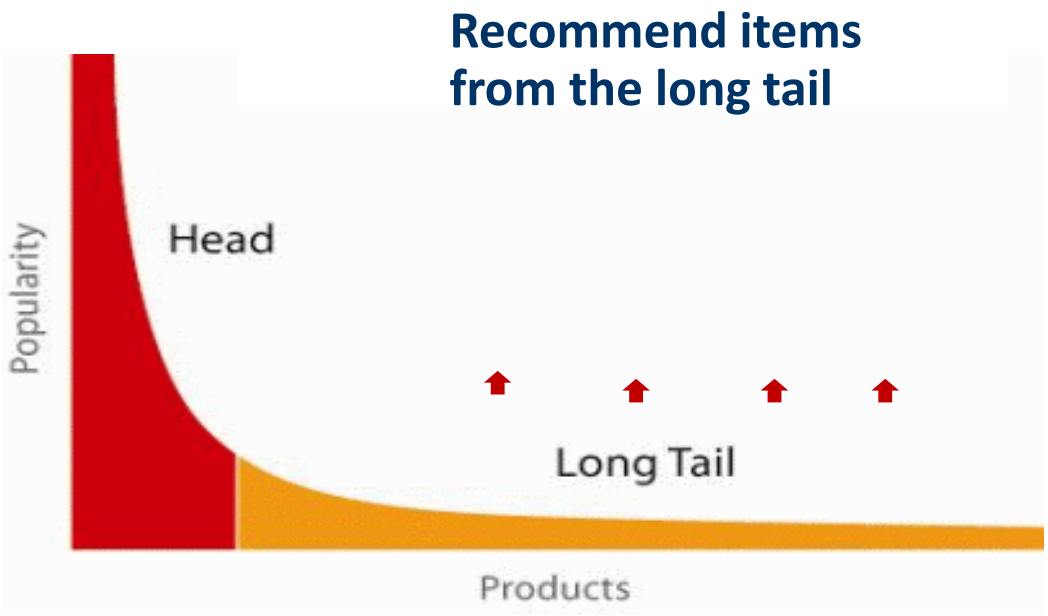
Different perspectives/aspects

- Depends on domain and purpose
- No holistic evaluation scenario exists

- **Retrieval perspective**
  - Reduce search costs
  - Provide "correct" proposals
  - Assumption: Users know in advance what they want

- **Recommendation perspective**
  - Serendipity – identify items from the Long Tail
  - Users did not know about existence

When does a RS do its job well?



- "Recommend widely unknown items that users might actually like!"
- 20% of items accumulate 74% of all positive ratings

Purpose and success criteria (2)

- **Prediction perspective**
  - Predict to what degree users like an item
  - Most popular evaluation scenario in research

- **Interaction perspective**
  - Give users a "good feeling"
  - Educate users about the product domain
  - Convince/persuade users - explain

- **Finally, conversion perspective**
  - Commercial situations
  - Increase "hit", "clickthrough", "lookers to bookers" rates
  - Optimize sales margins and profit

How do we as researchers know?

- **Test with real users**
  - A/B tests
  - Example measures: sales increase, click through rates
- **Laboratory studies**
  - Controlled experiments
  - Example measures: satisfaction with the system (questionnaires)
- **Offline experiments**
  - Based on historical data
  - Example measures: prediction accuracy, coverage

Empirical research

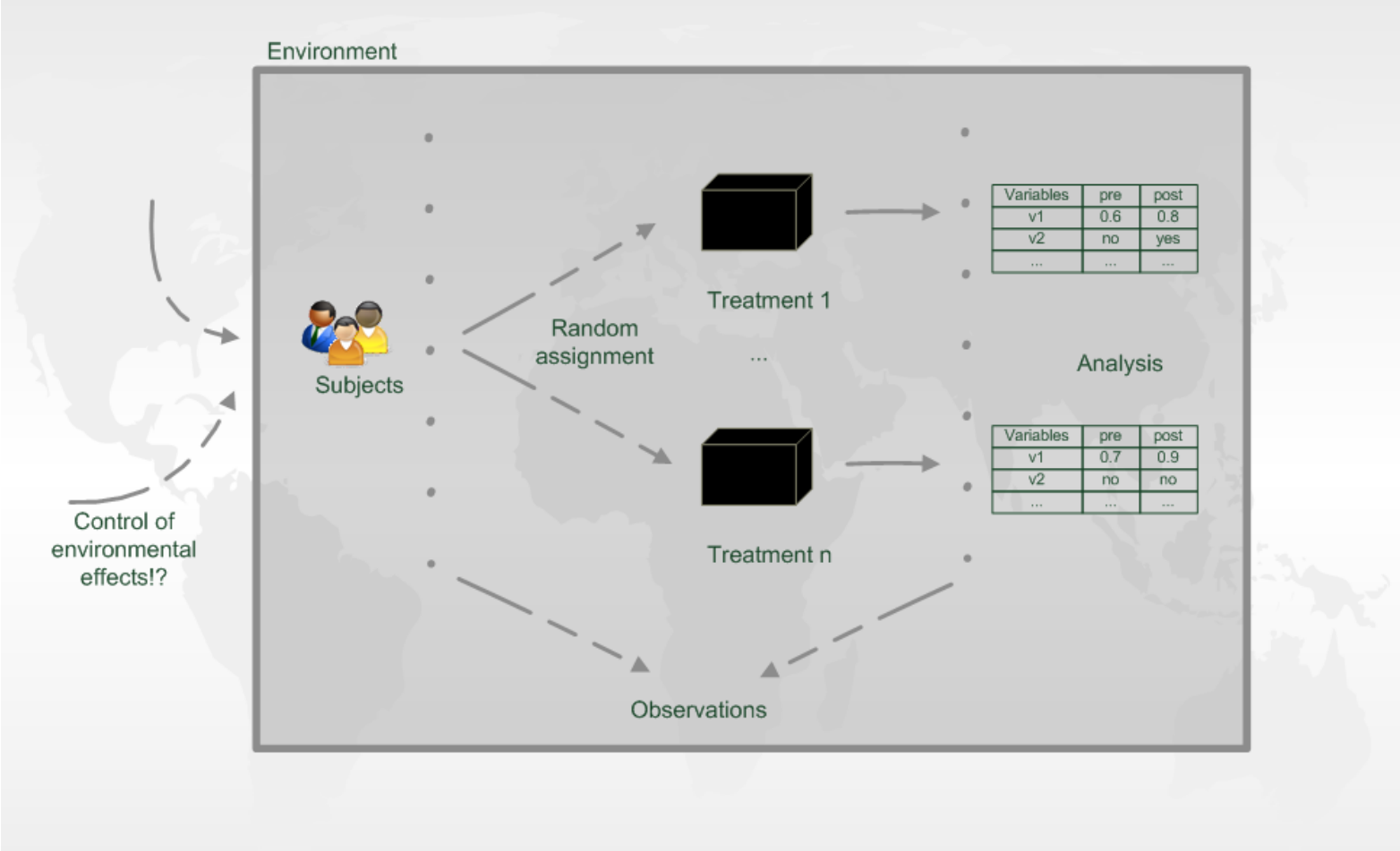
- **Characterizing dimensions:**
  - Who is the **subject** that is in the focus of research?
  - What **research method**s are applied?
  - In which **setting** does the research take place?

Subject	Online customers, students, historical online sessions, computers, …
Research method	Experiments, quasi-experiments, non-experimental research
Setting	Lab, real-world scenarios

Research methods

- **Experimental vs. non-experimental (observational) research methods**
  - Experiment (test, trial):
    - *"An experiment is a study in which at least one variable is manipulated and units are randomly assigned to different levels or categories of manipulated variable(s)."*
    - Units: users, historic sessions, …
    - Manipulated variable: type of RS, groups of recommended items, explanation strategies …
    - Categories of manipulated variable(s): content-based RS, collaborative RS

Experiment designs



Evaluation in information retrieval (IR)

- **Recommendation is viewed as information retrieval task:**
  - Retrieve (recommend) all items which are predicted to be "good" or "relevant".
- **Common protocol :**
  - Hide some items with known ground truth
  - Rank items or predict ratings -> Count -> Cross-validate
- **Ground truth established by human domain experts**

		Reality	
		Actually Good	Actually Bad
Prediction	Rated Good	True Positive (tp)	False Positive (fp)
	Rated Bad	False Negative (fn)	True Negative (tn)

Metrics: Precision and Recall

- **Precision: a measure of exactness, determines the fraction of relevant items retrieved out of all items retrieved**
  - E.g. the proportion of recommended movies that are actually good

$$Precision = \frac{tp}{tp + fp} = \frac{|good\ movies\ recommended|}{|\text{all recommendations}|}$$

- **Recall: a measure of completeness, determines the fraction of relevant items retrieved out of all relevant items**
  - E.g. the proportion of all good movies recommended

$$Recall = \frac{tp}{tp + fn} = \frac{|good\ movies\ recommended|}{|all\ good\ movies|}$$

Dilemma of IR measures in RS

- **IR measures are frequently applied, however**:
  - Ground truth for most items actually unknown
  - What is a relevant item?
  - Different ways of measuring precision possible
- Results from offline experimentation may have limited predictive power for online user behavior.

Metrics: Rank Score – position matters

For a user:

Actually good
Item 237
Item 899

hit

Recommended (predicted as good)
Item 345
Item 237
Item 187

- **Rank Score extends recall and precision to take the positions of correct items in a ranked list into account**
  - Particularly important in recommender systems as lower ranked items may be overlooked by users
  - Learning-to-rank: Optimize models for such measures (e.g., AUC)

Accuracy measures

- **Datasets with items rated by users**
  - MovieLens datasets 100K-10M ratings
  - Netflix 100M ratings
- **Historic user ratings constitute ground truth**
- **Metrics measure error rate**
  - Mean Absolute Error (*MAE*) computes the deviation between predicted ratings and actual ratings

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|p_i - r_i|$$

  - Root Mean Square Error (*RMSE*) is similar to *MAE*, but places more emphasis on larger deviation

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(p_i - r_i)^2}$$

Offline experimentation example

- **Netflix competition**
  - Web-based movie rental
  - Prize of $1,000,000 for accuracy improvement (RMSE) of 10% compared to own Cinematch system.

- **Historical dataset**
  - ~480K users rated ~18K movies on a scale of 1 to 5 (~100M ratings)
  - Last 9 ratings/user withheld
    - Probe set – for teams for evaluation
    - Quiz set – evaluates teams' submissions for leaderboard
    - Test set – used by Netflix to determine winner

- **Today**
  - Rating prediction only seen as an additional input into the recommendation process

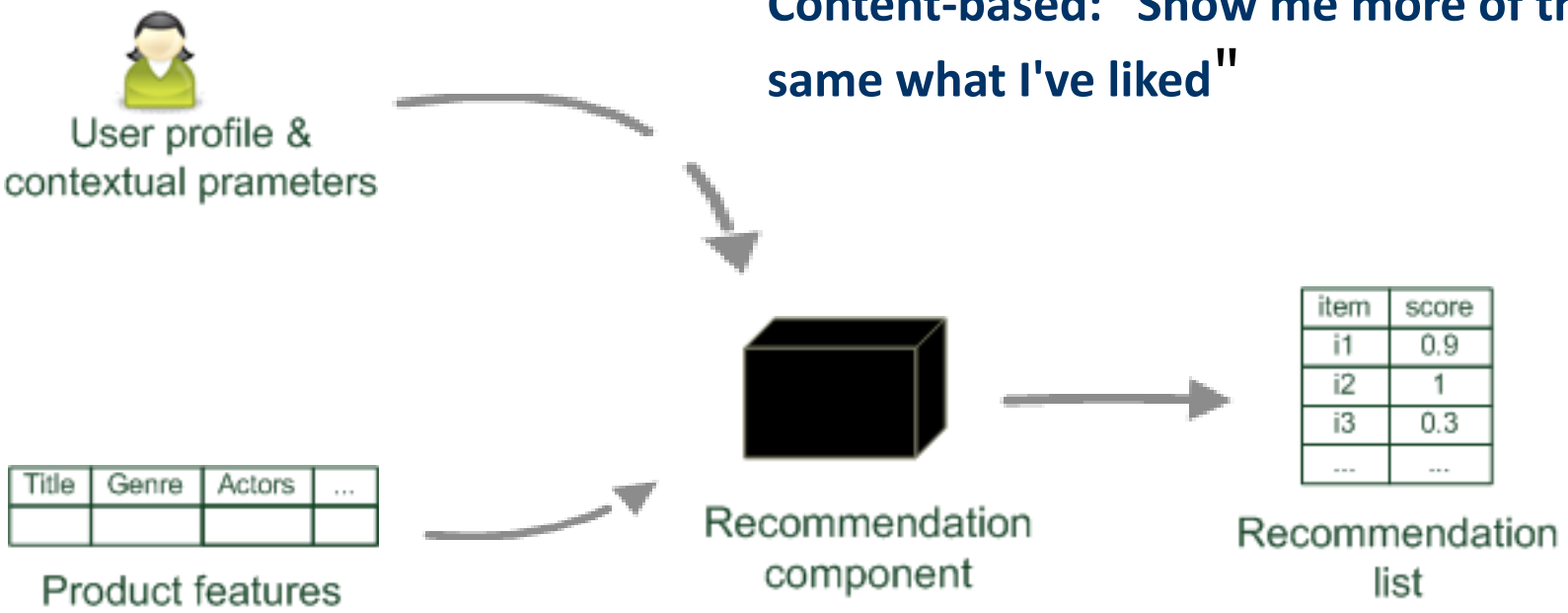
Content-based recommendation

Content-based recommendation

- **Collaborative filtering does NOT require any information about the items,**
  - However, it might be reasonable to exploit such information
  - E.g. recommend fantasy novels to people who liked fantasy novels in the past
- **What do we need:**
  - Some information about the available items such as the genre ("content")
  - Some sort of *user profile* describing what the user likes (the preferences)
- **The task:**
  - Learn user preferences
  - Locate/recommend items that are "similar" to the user preferences

Paradigms of recommender systems

Content-based: "Show me more of the same what I've liked"



What is the "content"?

- **The genre is actually not part of the content of a book**
- **Most CB-recommendation methods originate from Information Retrieval (IR) field:**
  - The item descriptions are usually automatically extracted (important words)
  - Goal is to find and rank interesting text documents (news articles, web pages)

- **Here:**
  - Classical IR-based methods based on keywords
  - No expert recommendation knowledge involved
  - User profile (preferences) are rather learned than explicitly elicited

Content representation and item similarities

Title	Genre	Author	Type	Price	Keywords
The Night of the Gun	Memoir	David Carr	Paperback	29.90	Press and journalism, drug addiction, personal memoirs, New York
The Lace Reader	Fiction, Mystery	Brunonia Barry	Hardcover	49.90	American contemporary fiction, detective, historical
Into the Fire	Romance, Suspense	Suzanne Brockmann	Hardcover	45.90	American fiction, Murder, Neo-nazism
...					

Title	Genre	Author	Type	Price	Keywords
...	Fiction, Suspense	Brunonia Barry, Ken Follet, ..	Paperback	25.65	detective, murder, New York

- **Simple approach**
  - Compute the similarity of an unseen item with the user profile based on the keyword overlap (e.g. using the Dice coefficient)
  - $\text{sim}(b_i, b_j) = \frac{2 * |keywords(b_i) \cap keywords(b_j)|}{|keywords(b_i)| + |keywords(b_j)|}$

Term-Frequency - Inverse Document Frequency (TF-IDF)

- **Simple keyword representation has its problems**
  - In particular when automatically extracted because
    - Not every word has similar importance
    - Longer documents have a higher chance to have an overlap with the user profile
- **Standard measure: TF-IDF**
  - Encodes text documents as weighted term vector
  - TF: Measures, how often a term appears (density in a document)
    - Assuming that important terms appear more often
    - Normalization has to be done in order to take document length into account
  - IDF: Aims to reduce the weight of terms that appear in all documents

TF-IDF

- **Compute the overall importance of keywords**
  - Given a keyword i and a document j

    $$TF\text{-}IDF\ (i,j) = TF(i,j) * IDF(i)$$

- **Term frequency (TF)**
  - Let $freq(i,j)$ number of occurrences of keyword *i* in document *j*
  - Let $maxOthers(i,j)$ denote the highest number of occurrences of another keyword of *j*
  - $TF(i,j) = \frac{freq(i,j)}{maxOthers(i,j)}$

- **Inverse Document Frequency (IDF)**
  - N: number of all recommendable documents
  - n(i): number of documents in which keyword *i* appears
  - $IDF(i) = log\frac{N}{n(i)}$

Example TF-IDF representation

	Antony and Cleopatra	Julius Caesar	The Tempest	Hamlet	Othello	Macbeth
Antony	5.25	3.18	0	0	0	0.35
Brutus	1.21	6.1	0	1	0	0
Caesar	8.59	2.54	0	1.51	0.25	0
Calpurnia	0	1.54	0	0	0	0
Cleopatra	2.85	0	0	0	0	0
mercy	1.51	0	1.9	0.12	5.25	0.88
worser	1.37	0	0.11	4.15	0.25	1.95

Figure taken from http://informationretrieval.org

More on the vector space model

- **Vectors are usually long and sparse**
- **Improvements**
  - Remove stop words ("a", "the", ..)
  - Use stemming
  - Size cut-offs (only use top n most representative words, e.g. around 100)
  - Use additional knowledge, use more elaborate methods for feature selection
  - Detection of phrases as terms (such as United Nations)
- **Limitations**
  - Semantic meaning remains unknown
  - Example: usage of a word in a negative context
    - "there is **nothing** on the menu that a vegetarian would like.."
- **Usual similarity metric to compare vectors: Cosine similarity (angle)**

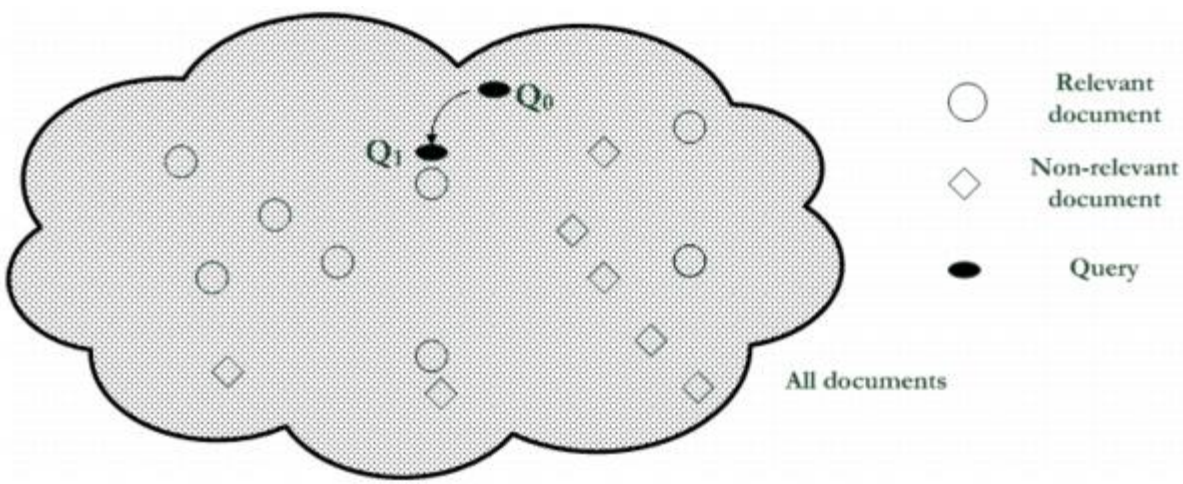
Recommending items

- **Simple method: nearest neighbors**
  - Given a set of documents D already rated by the user (like/dislike)
    - Find the n nearest neighbors of a not-yet-seen item *i* in D
    - Take these ratings to predict a rating/vote for *i*
    - (Variations: neighborhood size, lower/upper similarity thresholds)
- **Query-based retrieval: Rocchio's method**
  - The SMART System: Users are allowed to rate (relevant/irrelevant) retrieved documents (feedback)
  - The system then learns a prototype of relevant/irrelevant documents
  - Queries are then automatically extended with additional terms/weight of relevant documents

Rocchio details

- Document collections D^+ and D^-
- α, β, γ used to fine-tune the feedback
- often only positive feedback is used

$$Q_{i+1} = \alpha * Q_i + \beta\left(\frac{1}{|D^+|}\sum_{d^+ \in D^+} d^+\right) - \gamma\left(\frac{1}{|D^-|}\sum_{d^- \in D^-} d^-\right)$$



Probabilistic methods

- **Recommendation as classical text classification problem**
  - Long history of using probabilistic methods
- **Simple approach:**
  - 2 classes: like/dislike
  - Simple Boolean document representation
  - Calculate probability that document is liked/disliked based on Bayes theorem

Doc-ID	recommender	intelligent	learning	school	**Label**
1	1	1	1	0	**1**
2	0	0	1	1	**0**
3	1	1	0	0	**1**
4	1	0	1	1	**1**
5	0	0	0	1	**0**
6	1	1	0	0	**?**

Remember:
P(Label=1|X)=
k*P(X|Label=1) * P(Label=1)

$$
\begin{aligned}
P(X|Label{=}1) &= P(recommender{=}1|Label{=}1) \times \\
&\quad P(intelligent{=}1|Label{=}1) \times \\
&\quad P(learning{=}0|Label{=}1) \times P(school{=}0|Label{=}1) \\
&= 3/3 \times 2/3 \times 1/3 \times 2/3 \\
&\approx 0.149
\end{aligned}
$$

Improvements

- **Side note: Conditional independence of events does in fact not hold**
  - “New”/ “York“ and “Hong” / “Kong"
  - Still, good accuracy can be achieved
- **Boolean representation simplistic**
  - Keyword counts lost
- **More elaborate probabilistic methods**
  - E.g. estimate probability of term **v** occurring in a document of class **C** by relative frequency of **v** in all documents of the class
- **Other linear classification algorithms (machine learning) can be used**
  - Support Vector Machines, ..

Limitations of content-based recommendation methods

- **Keywords alone may not be sufficient to judge quality/relevance of a document or web page**
  - Up-to-dateness, usability, aesthetics, writing style
  - Content may also be limited / too short
  - Content may not be automatically extractable (multimedia)
- **Ramp-up phase required**
  - Some training data is still required
  - Web 2.0: Use other sources to learn the user preferences
- **Overspecialization**
  - Algorithms tend to propose "more of the same"
  - E.g. too similar news items

Knowledge-Based Recommender Systems

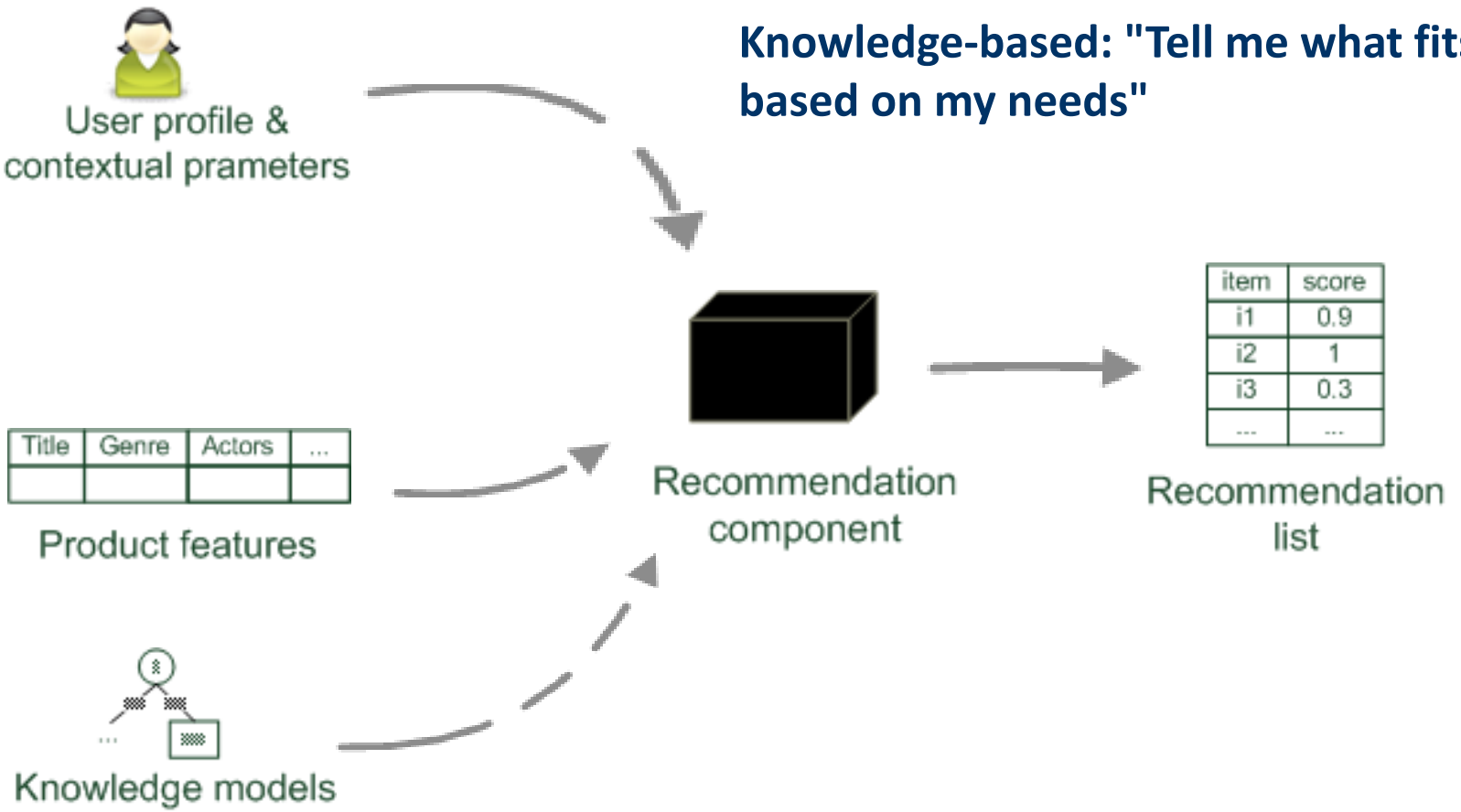
Why do we need knowledge-based recommendation?

- **Products with low number of available ratings**

- **Time span plays an important role**
  - Five-year-old ratings for computers
  - User lifestyle or family situation changes
- **Customers want to define their requirements explicitly**
  - “The color of the car should be black"

Knowledge-based recommendation

Knowledge-based: "Tell me what fits based on my needs"



Knowledge-based recommendation I

- **Explicit domain knowledge**
  - Sales knowledge elicitation from domain experts
  - System mimics the behavior of experienced sales assistant
  - Best-practice sales interactions
  - Can guarantee “correct” recommendations (determinism) with respect to expert knowledge

- **Conversational interaction strategy**
  - Opposed to one-shot interaction
  - Elicitation of user requirements
  - Transfer of product knowledge (“educating users”)

Knowledge-Based Recommendation II

- **Different views on “knowledge”**
  - Similarity functions
    - Determine matching degree between query and item (case-based RS)
  - Utility-based RS
    - E.g. MAUT – Multi-attribute utility theory
  - Logic-based knowledge descriptions (from domain expert)
    - E.g. Hard and soft constraints

Constraint-based recommendation I

- **A knowledge-based RS formulated as constraint satisfaction problem**

$$CSP(X_I \cup X_U, D, SRS \cup KB \cup I)$$

- **Def.**
  - X_I, X_U: Variables describing items and user model with domain D (e.g. lower focal length, purpose)
  - KB: Knowledge base comprising constraints and domain restrictions (e.g. **IF** purpose="*on travel*" **THEN** lower focal length < *28mm*)
  - SRS: Specific requirements of a user (e.g. purpose = "*on travel*")
  - I: Product catalog (e.g. (id=1 ∧ lfl = *28mm)* ∨ (id=2 ∧ lfl= *35mm*) ∨ …)
- **Solution: Assignment tuple θ assigning values to all variables X_I**

$s.t.\ SRS \cup KB \cup I \cup \theta$ **is satisfiable.**

Item ranking

- **Multi-Attribute Utility Theory (MAUT)**
  - Each item is evaluated according to a predefined set of dimensions that provide an aggregated view on the basic item properties
- **E.g. <u>quality</u> and <u>economy</u> are dimensions in the domain of digital cameras**

id	value	quality	economy
price	≤250 >250	5 10	10 5
mpix	≤8 >8	4 10	10 6
opt-zoom	≤9 >9	6 10	9 6
…	…	…	…

Customer-specific item utilities with MAUT

- **Customer interests:**

customer	quality	economy
Cu_1	80%	20%
Cu_2	40%	60%

*

- **Item utilities:**

quality	economy	utility: cu_1	utility: cu_2
P1 Σ(5,4,6,6,3,7,10) = 41	Σ (10,10,9,10,10,10,6) = 65	45.8 [8]	55.4 [6]
P2 Σ(5,4,6,6,10,10,8) = 49	Σ (10,10,9,10,7,8,10) = 64	52.0 [7]	58.0 [1]
P3 Σ(5,4,10,6,10,10,8) = 53	Σ (10,10,6,10,7,8,10) = 61	54.6 [5]	57.8 [2]
...	...	...	...

**

$$utility(p) = \sum_{j=1}^{\#(dimensions)} \underbrace{interest(j)}_{*} * \underbrace{contribution(p,j)}_{**}$$

Constraint-based recommendation II

- **BUT: What if no solution exists?**
  - $KB \cup I$ not satisfiable → debugging of knowledge base
  - $SRS \cup KB \cup I$ not satisfiable but $KB \cup I$ correct → debugging of user requirements

- **Application of model-based diagnosis for debugging user requirements**
  - Diagnoses: $(SRS \setminus \Delta) \cup KB \cup I$ is satisfiable
  - Repairs: $(SRS \setminus \Delta) \cup \Delta_{repair} \cup KB \cup I$ is satisfiable
  - Conflict sets: $CS \subseteq SRS : CS \cup KB \cup I$ not satisfiable

Example: find minimal relaxations (minimal diagnoses)

Knowledge Base:

	LHS	RHS
C1	TRUE	Brand = Brand pref.
C2	Motives = *Landscape*	Low. foc. Length =< *28*
C3	TRUE	Price =< Max. cost

Product catalogue:

Powershot XY	
Brand	*Canon*
Lower focal length	*35*
Upper focal length	*140*
Price	*420 EUR*

Lumix	
Brand	*Panasonic*
Lower focal length	*28*
Upper focal length	*112*
Price	*319 EUR*

Current user:

		User model (SRS)	
CS1	R1	Motives	*Landscape*
	R2	Brand preference	*Canon*
CS2	R3	Max. cost	*350 EUR*

Diagnoses: $\Delta_1 = \{R2\}, \Delta_2 = \{R1, R3\}$

Ask user

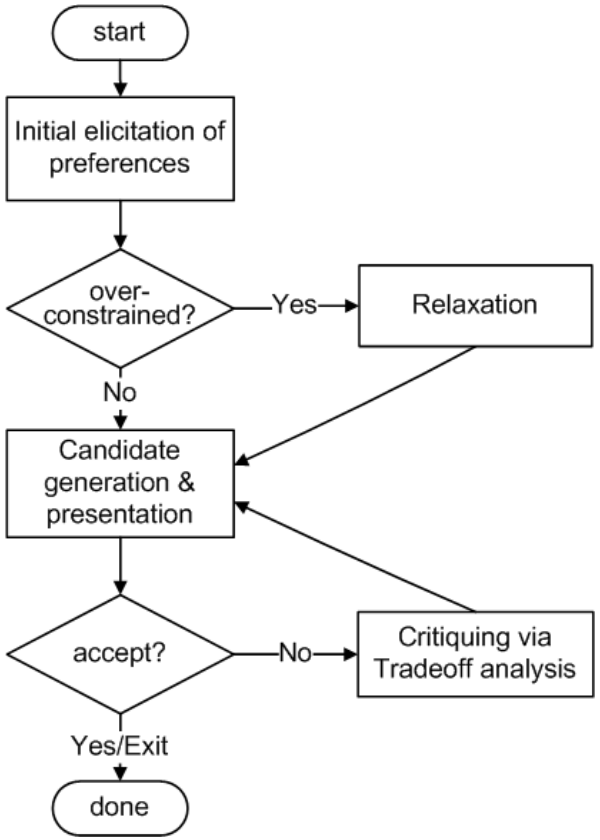
- **Computation of minimal revisions of requirements**
  - Do you want to relax your brand preference?
    - Accept *Panasonic* instead of *Canon* brand
  - Or is photographing landscapes with a wide-angle lens and maximum cost less important?
    - Lower focal length > 28mm and Price > 350 EUR
  - Optionally guided by some predefined weights or past community behavior
- **Be aware of possible revisions** (e.g. age, family status, …)

Constraint-based recommendation III

- **More variants of recommendation task**
  - Customers maybe not know what they are seeking
  - Find "diverse" sets of items
    - Notion of similarity/dissimilarity
    - Idea that users navigate a product space
    - If recommendations are more diverse than users can navigate via critiques on recommended "entry points" more efficiently (less steps of interaction)
  - Bundling of recommendations
    - Find item bundles that match together according to some knowledge
      - E.g. travel packages, skin care treatments or financial portfolios
      - RS for different item categories, CSP restricts configuring of bundles

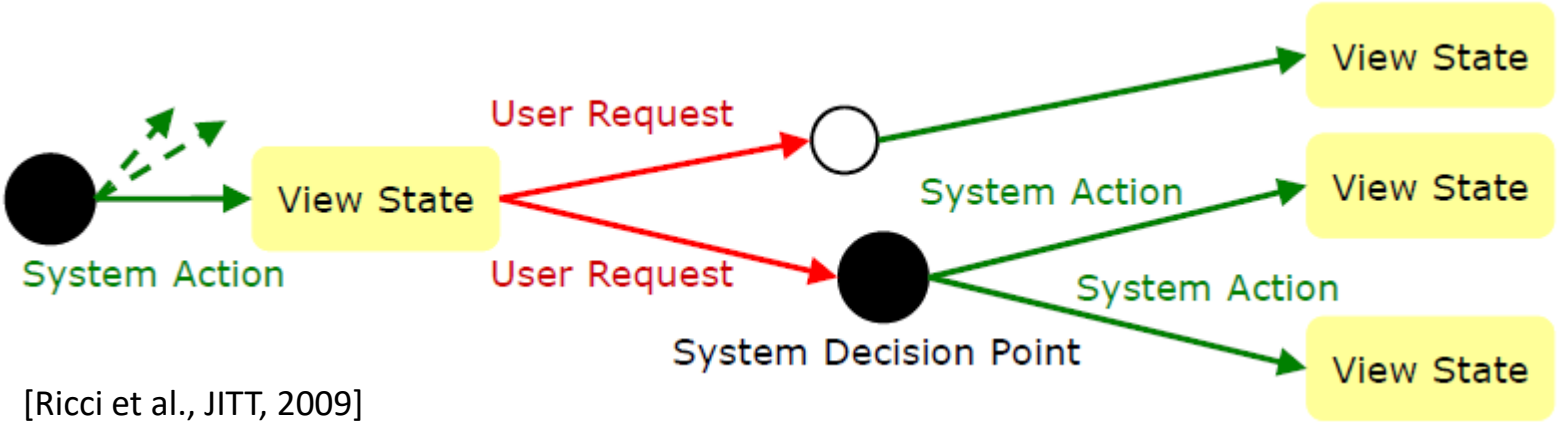
Conversational strategies

- **Process consisting of multiple conversational moves**
  - Resembles natural sales interactions
  - Not all user requirements known beforehand
  - Customers are rarely satisfied with the initial recommendations
- **Different styles of preference elicitation:**
  - Free text query interface
  - Asking technical/generic properties
  - Images / inspiration
  - Proposing and Critiquing



Example: adaptive strategy selection

- **State model, different actions possible**
  - Propose item, ask user, relax/tighten result set,…



[Ricci et al., JITT, 2009]

Limitations of knowledge-based recommendation methods

- **Cost of knowledge acquisition**
  - From domain experts
  - From users
  - Remedy: exploit web resources

- **Accuracy of preference models**
  - Very fine granular preference models require many interaction cycles with the user or sufficient detailed data about the user
  - Remedy: use collaborative filtering, estimates the preference of a user

  However: preference models may be instable

  - E.g. asymmetric dominance effects and decoy items